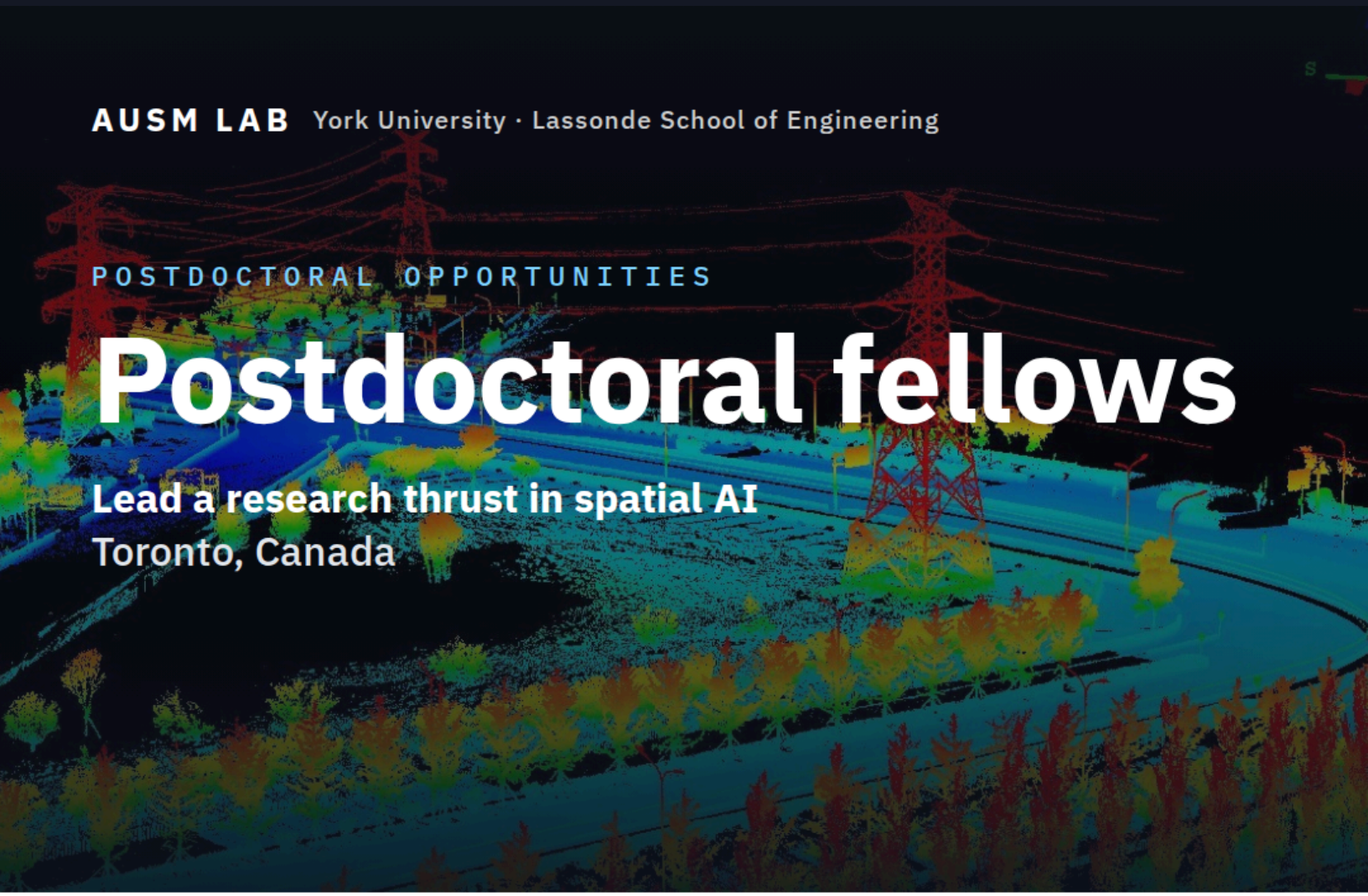
AUSM LAB York University · Lassonde School of Engineering



POSTDOCTORAL OPPORTUNITIES

Postdoctoral fellows

Lead a research thrust in spatial AI

Toronto, Canada

Infrastructure AI

Embodied spatial AI for autonomous infrastructure monitoring (NSERC Discovery)

Aerial Digital Twins

Drone LiDAR and imagery, LoD2 and BIM models with industry partners

Space Awareness

Resident space object detection and tracking from on-orbit sensors

Start
Flexible (2026–2027)

Applications
Reviewed on a rolling basis

Email your **CV, research statement and 2–3 papers** to **gsohn@yorku.ca**
Subject: **[Postdoc] Your Name**

ausmlab.com/join

POSTDOCTORAL OPPORTUNITIES

Postdoctoral fellows

Applications reviewed on a rolling basis · Start: Flexible (2026–2027)

Lead research on structured spatial world models, mentor graduate students, and work directly with our industry partners on real-world sensing and mapping problems.

ABOUT THE LAB

The AUSM Lab at York University develops embodied spatial intelligence: AI systems that perceive, reason about and act within physical space. Our work spans autonomous infrastructure monitoring, aerial mapping and digital twins, smart mobility, and AI for Earth and space observation.

We are looking for a postdoctoral fellow to lead one of our research thrusts. You will work at the centre of a team of Ph.D. and M.Sc. students, and with industry and government partners such as Teledyne Optech, Aethon Aerial Solutions and the Ontario Ministry of Transportation.

RESEARCH THRUSTS

- **Infrastructure AI**
  Embodied spatial AI for autonomous infrastructure monitoring (NSERC Discovery)
- **Aerial Digital Twins**
  Drone LiDAR and imagery, LoD2 and BIM models with industry partners
- **Space Awareness**
  Resident space object detection and tracking from on-orbit sensors

WE OFFER

- Funding through NSERC Discovery, NSERC CREATE SMART and industry partners
- Support to apply for the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate fellowships
- Experience leading industry projects and mentoring graduate students
- A GPU computing cluster and real-world data from industry partners

YOUR ROLE

- Lead a research thrust, from problem definition to publication
- Publish in top venues (CVPR, ICCV, ECCV, ISPRS, IEEE TGRS)
- Co-supervise Ph.D. and M.Sc. students day to day
- Run industry projects and meet with partners
- Help write grant and fellowship proposals

WHO WE LOOK FOR

- A Ph.D. (completed or near completion) in computer vision, machine learning, robotics, photogrammetry and remote sensing, geomatics, or a related field.
- A strong publication record in leading venues (e.g., CVPR, ICCV, ECCV, ISPRS, IEEE TGRS).
- Hands-on experience with deep learning (Python, PyTorch) and 3D or spatial data.

HOW TO APPLY

Email **Prof. Gunho Sohn, gsohn@yorku.ca** with:

- Your CV with a full publication list
- A research statement (up to two pages)
- Two or three representative publications
- Names and contact details of three referees

Subject line: **[Postdoc] Your Name**

FUNDING

Positions are funded through the lab's research grants and industry partners. We also encourage strong candidates to apply with us for external fellowships, such as the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate, and we support the application.